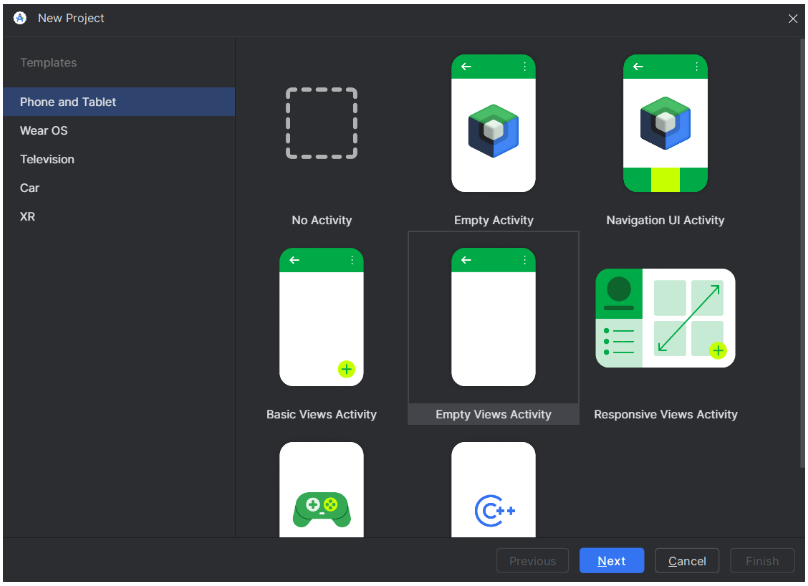
3. Kotlin Basics Demonstration App

Topics covered: variables, data types, type conversion, operators, control statements (if/when), loops
Task:

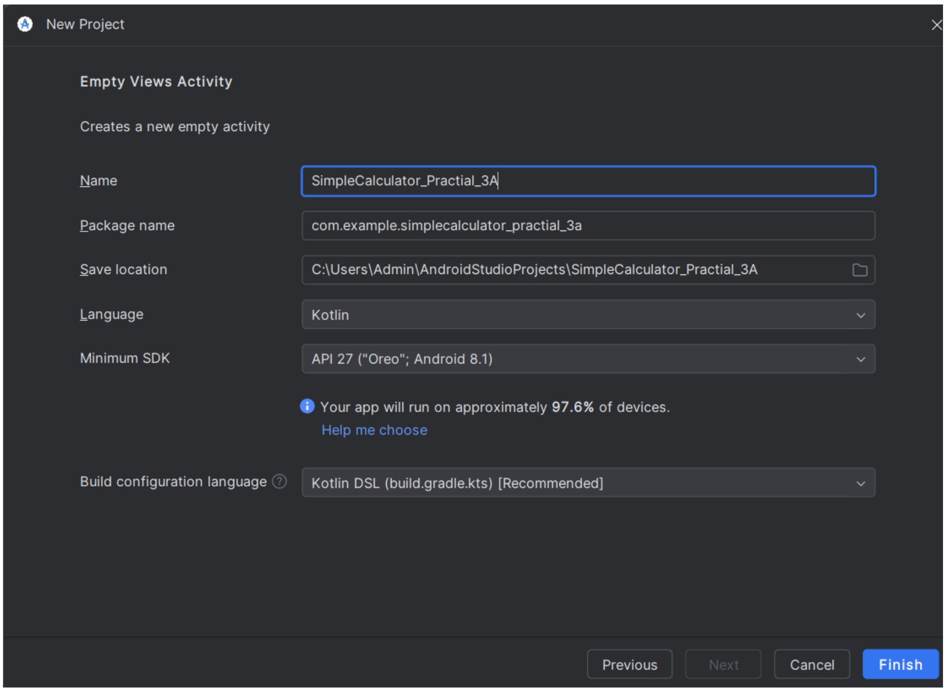
3A. Take two numbers as input. Perform addition, subtraction, multiplication. Use `if` and `when` conditions to show results and show result in Logcat

Step 1:

Go to File→ New→ New Project→Empty Views Activity→ Next



Click on Project name as “SimpleCalculator_Practial_3A” → Finish



Step 3: Open the activity_main.xml

1. Go to
   app → src → main → res → layout → activity_main.xml
2. Click it.

You will see **Design / Code / Split** tabs at the top.

Click **Split** (so you can see both code + preview) or Click (Alt+Shift+Left). You will see xml code.

Replace it with this given code:

```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp">

    <EditText
        android:id="@+id/etNum1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter First Number"
        android:inputType="numberDecimal" />

    <EditText
        android:id="@+id/etNum2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Second Number"
        android:inputType="numberDecimal"
        android:layout_below="@id/etNum1"
        android:layout_marginTop="15dp" />

    <Button
        android:id="@+id/btnAdd"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Add"
        android:layout_below="@id/etNum2"
        android:layout_marginTop="20dp" />

    <Button
        android:id="@+id/btnSub"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Subtract"
        android:layout_toRightOf="@id/btnAdd"
        android:layout_alignTop="@id/btnAdd"
        android:layout_marginLeft="20dp" />

    <Button
        android:id="@+id/btnMul"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Multiply"
        android:layout_toRightOf="@id/btnSub"
        android:layout_alignTop="@id/btnAdd"
```

```
        android:layout_marginLeft="20dp" />

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Result will appear here"
        android:textSize="20sp"
        android:layout_below="@id/btnAdd"
        android:layout_marginTop="30dp" />

</RelativeLayout>
```

Step 4: **Open MainActivity.kt**

Find this file:

app → src → main → java → com.example.SimpleCalculator_Practial_3A → MainActivity.kt

Open it.

Add the import at first:

```
import android.widget.Button
import android.widget.EditText
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity
import android.util.Log
```

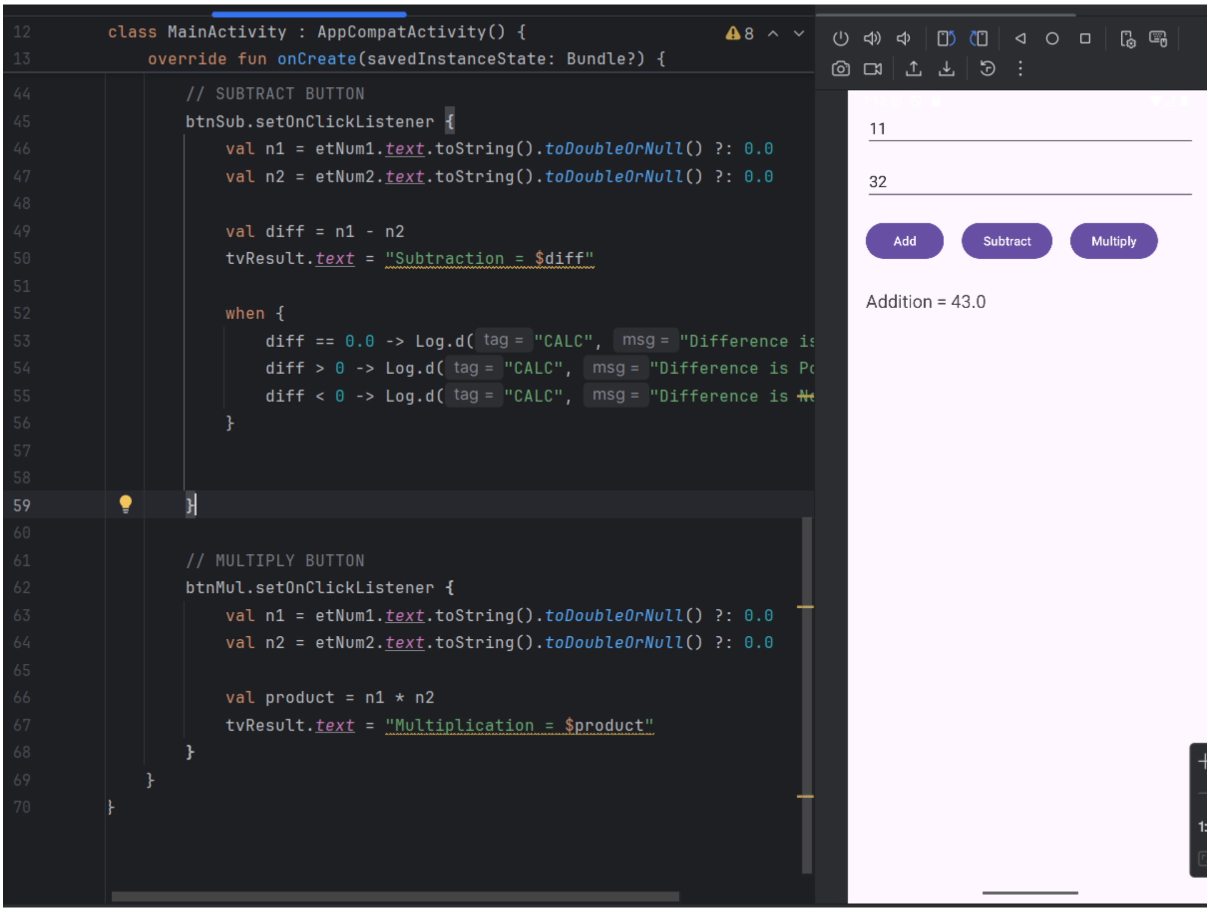
Then replace this:

Add this code:

```
class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        // Get views from XML
        val etNum1 = findViewById<EditText>(R.id.etNum1)
        val etNum2 = findViewById<EditText>(R.id.etNum2)
        val btnAdd = findViewById<Button>(R.id.btnAdd)
        val btnSub = findViewById<Button>(R.id.btnSub)
        val btnMul = findViewById<Button>(R.id.btnMul)
        val tvResult = findViewById<TextView>(R.id.tvResult)

        // ADD BUTTON
        btnAdd.setOnClickListener {
            val n1 = etNum1.text.toString().toDoubleOrNull() ?: 0.0
            val n2 = etNum2.text.toString().toDoubleOrNull() ?: 0.0

            val sum = n1 + n2
            tvResult.text = "Addition = $sum"

            if (sum > 0) {
                Log.d("CALC", "Sum is Positive")
            } else if (sum < 0) {
```

```kotlin
                Log.d("CALC", "Sum is Negative")
            } else {
                Log.d("CALC", "Sum is Zero")
            }


        }

        // SUBTRACT BUTTON
        btnSub.setOnClickListener {
            val n1 = etNum1.text.toString().toDoubleOrNull() ?: 0.0
            val n2 = etNum2.text.toString().toDoubleOrNull() ?: 0.0

            val diff = n1 - n2
            tvResult.text = "Subtraction = $diff"

            when {
                diff == 0.0 -> Log.d("CALC", "Difference is Zero")
                diff > 0 -> Log.d("CALC", "Difference is Positive")
                diff < 0 -> Log.d("CALC", "Difference is Negative")
            }


        }

        // MULTIPLY BUTTON
        btnMul.setOnClickListener {
            val n1 = etNum1.text.toString().toDoubleOrNull() ?: 0.0
            val n2 = etNum2.text.toString().toDoubleOrNull() ?: 0.0

            val product = n1 * n2
            tvResult.text = "Multiplication = $product"
        }
    }
}
```

Output (Virtual Phone):



Step 6: Connection this app in Actual phone (your phone)

1. Open your YOUR phone

Unlock it.

2. Open the "Settings" app

It's a **gear** ⚙️ **icon** (on your home screen or app drawer).

3. Scroll and tap "About device"

On OPPO, it is usually at the **top** of settings.

4. Tap "Version"

Inside About Device.

5. Tap "Build number" 7 times

Your phone will say:
"You are now a developer!"

6. Go back → Additional Settings

Scroll and find:
Additional Settings OR

7. Search and Tap "Developer Options"

Now you will see:
Developer Options (only appears after enabling it)

8. Turn ON USB Debugging

Scroll down and enable:

➡️ **USB Debugging**
Tap **OK** to allow it.

Now connect you're YOUR phone to your laptop

Use your USB cable → as soon as you connect, your phone will show:

"Allow USB debugging?"

Tap:

✓ Always allow from this computer
✓ Allow

✓ USB Mode MUST be "File Transfer"

Once your Phone is connected

✓ Select your phone:

At the top device dropdown, choose for example:

👉 **OPPO CPH2557**

(It should show a green phone icon)

✓ Then click the green RUN ▶️ button.

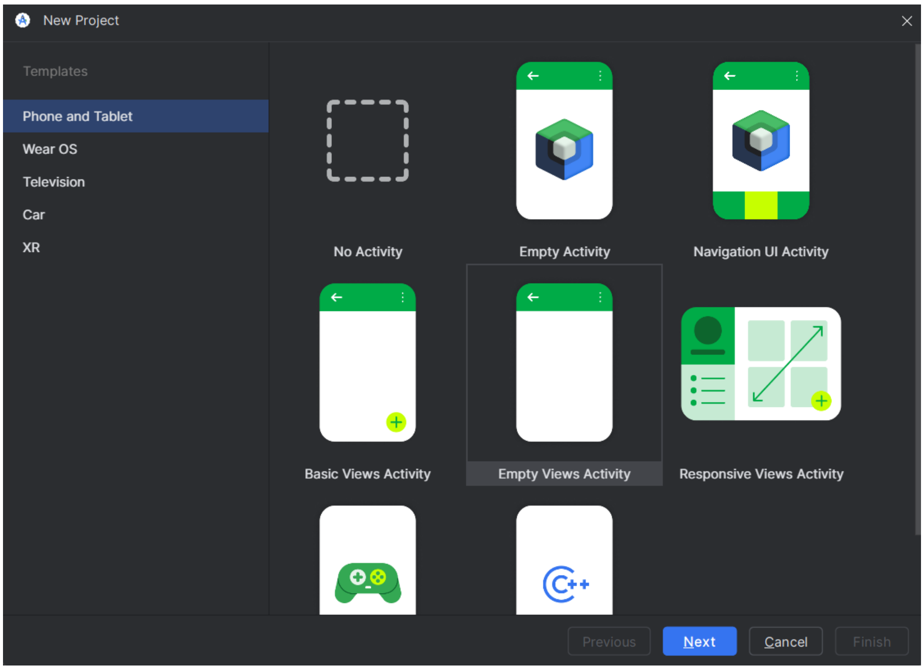
Your app will install and start on your phone.

Output (Real phone):

3B. Print loop output (1 to 10) in Logcat

Step 1:

Go to File→ New→ New Project→Empty Views Activity→ Next



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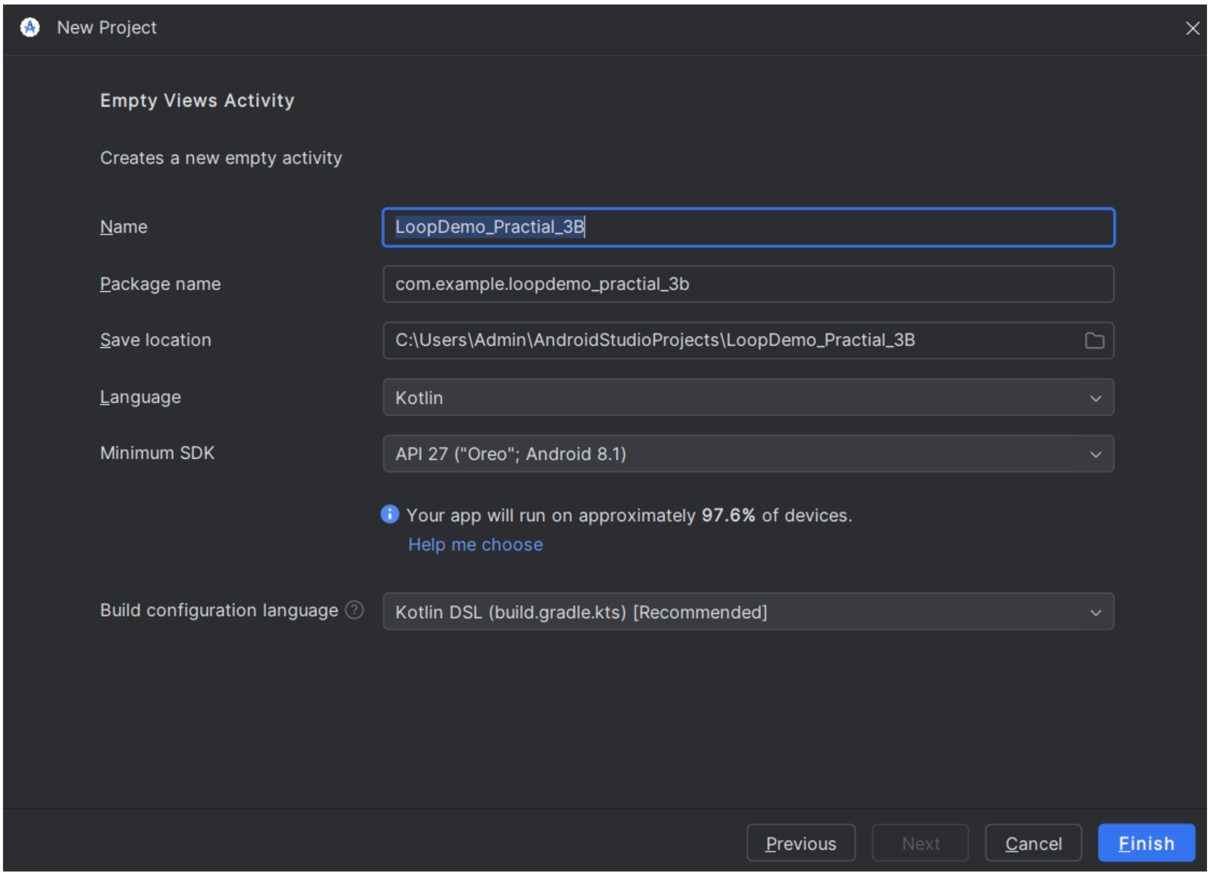

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    android:layout_width="match_parent"
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    android:padding="20dp">

    <Button
        android:id="@+id/btnPrintLoop"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Print 1..10"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="50dp"/>

    <TextView
        android:id="@+id/tvLoopOutput"
```

```
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Loop output will appear in Logcat"
        android:layout_below="@id/btnPrintLoop"
        android:layout_marginTop="30dp"
        android:textSize="18sp"/>

</RelativeLayout>
```

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import android.util.Log
```

Then replace this:

Add this code:

```
class MainActivity : AppCompatActivity() {

    private val TAG = "LOOP"   // Tag for Logcat

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val btnPrintLoop = findViewById<Button>(R.id.btnPrintLoop)

        btnPrintLoop.setOnClickListener {

            // For loop 1..10
            for (i in 1..10) {
                Log.d(TAG, "For loop value: $i")
            }

            // While loop 1..10
            var j = 1
            while (j <= 10) {
                Log.d(TAG, "While loop value: $j")
                j++
            }

            // Do-while loop 1..10
            var k = 1
            do {
                Log.d(TAG, "Do-While loop value: $k")
                k++
            } while (k <= 10)
        }
```

```
    }
}
```

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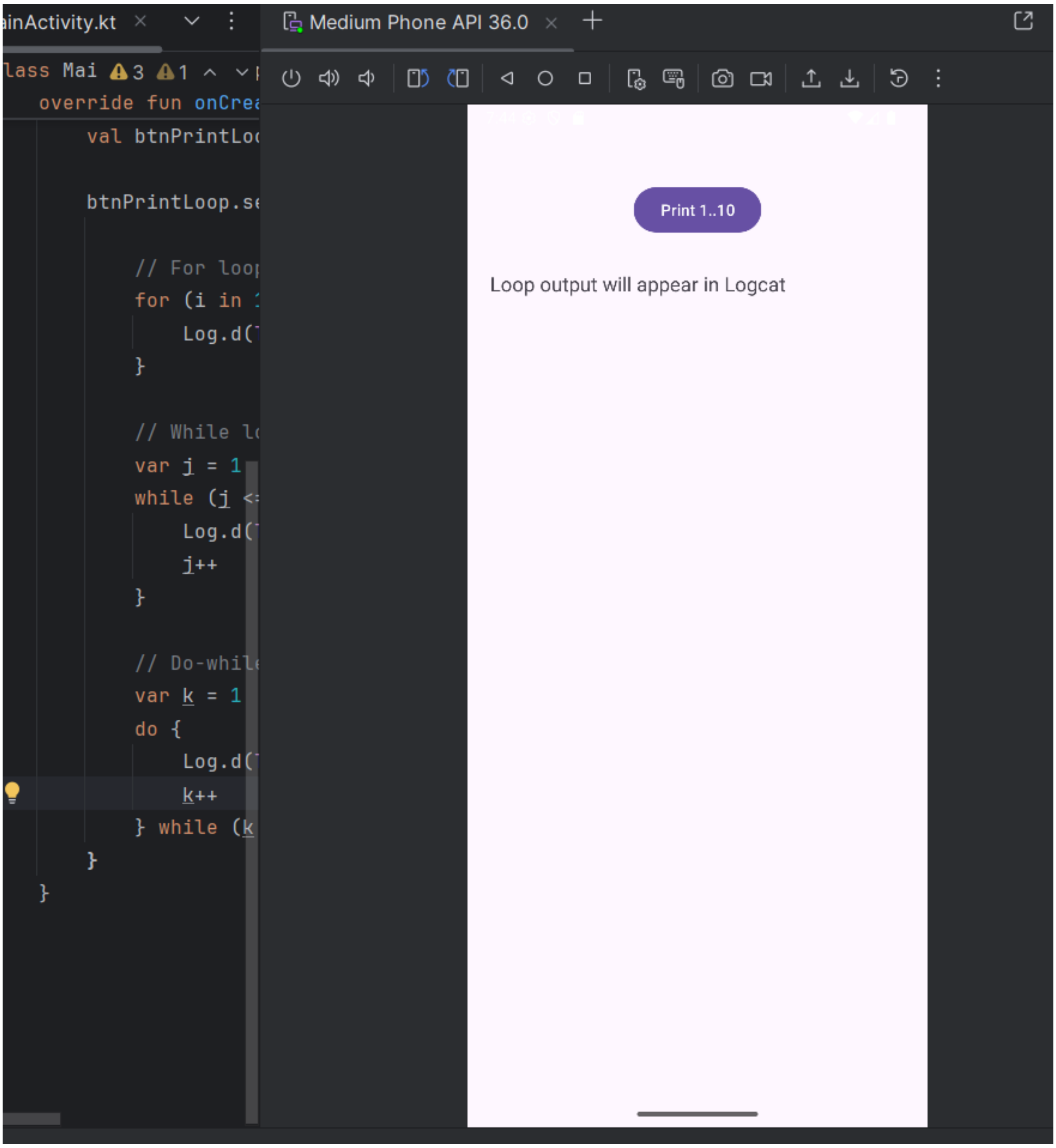

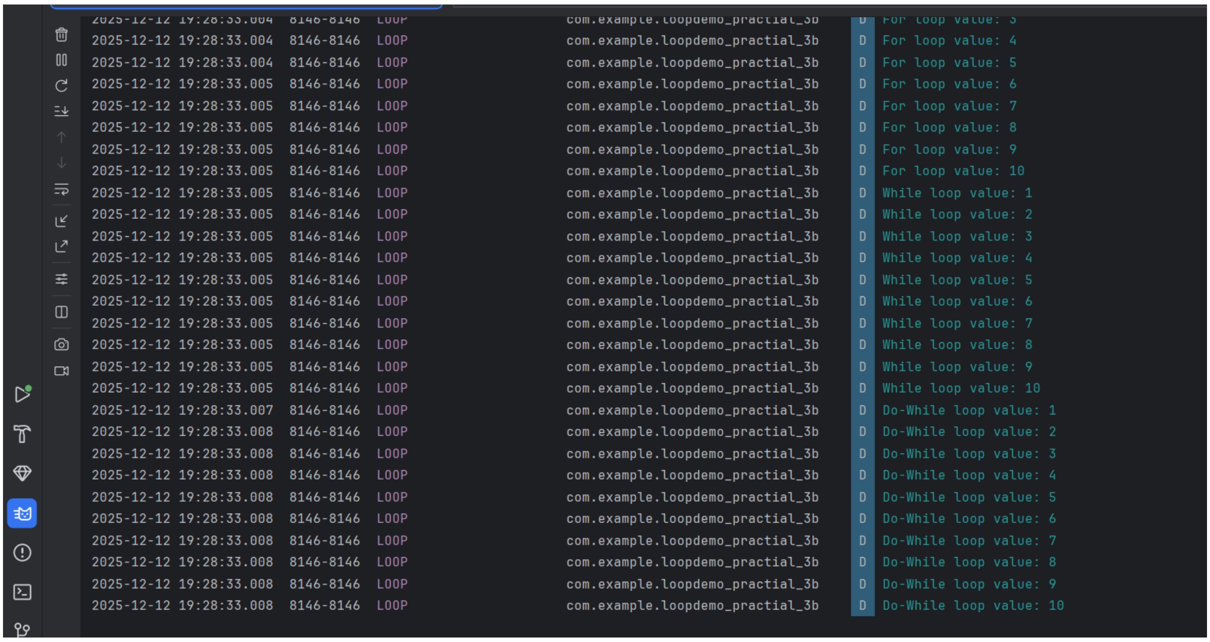

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